

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

15

10 answers · Rationale-rich · Teach from doc

Q1**C**

- C. the period of interbreeding resulted in a greater genetic difference between alpacas and their wild ancestors than between llamas and their wild ancestors.

Choice C is the best answer because it presents the inference that most logically follows from the text's discussion of the genetic material of alpacas. The text states that modern alpacas get only 64 percent of their genetic material from their wild ancestor (the vicuña), while llamas get 95.5 percent of their genetic material from their wild ancestor (the guanaco). These data imply that relative to llamas, alpacas have experienced a much greater genetic shift away from their wild ancestors (36 percent of their genetic material derives from another source, compared to 4.5 percent of llamas' genetic material). Moreover, the text establishes that the remainder of alpacas' genetic material derives from llamas, and the remainder of llamas' genetic material derives from alpacas, reflecting a period of interbreeding. Taken together, these details support the inference that the period of interbreeding had a more significant impact on the genetic material of alpacas than on that of llamas, thereby reducing the percentage of alpacas' genetic material that derives from wild ancestors and placing alpacas at a greater remove from these ancestors.

Choice A is incorrect because the text doesn't provide information about genetic diversity within modern llama and alpaca populations. While the text discusses the percentage of genetic material each domesticated species inherited from its wild ancestor and from interbreeding, it does not address the level of genetic variation within each species. Therefore, it isn't logical to infer that modern llama populations have a greater degree of genetic diversity than modern alpaca populations do. Choice B is incorrect because it contradicts the chronology presented in the text. The text states that alpacas were domesticated about 7,000 years ago, but the interbreeding with llamas occurred "between 400 and 600 years ago." The genetic mixing happened thousands of years after domestication, not during it; in addition, the text does not provide details about what the process of domestication might entail. Therefore, it isn't logical to infer that llama genetic material was introduced during the domestication process of alpacas. Choice D is incorrect because though the text notes that guanacos and vicuñas contributed to the genetic material of alpacas and llamas, it only discusses alpacas' and llamas' genetic composition. Therefore, the text does not support the inference that modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding.

Q2

A

- A. When plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid and more salicylic acid than did control plants.

Choice A is the best answer because it best describes data from the graph that support the researchers' conclusion about whitefly-induced defenses in tomato plants. According to the text, tomato plants produce airborne chemicals that prompt nearby tomato plants to increase their production of appropriate defensive chemicals, such as jasmonic acid in response to herbivory and salicylic acid in response to microbial infection. The text goes on to indicate that researchers concluded that whiteflies induce tomato plants to emit chemicals that cause nearby plants to defend against microbial infection instead of herbivory from whiteflies. The graph shows that control tomato plants (plants that weren't exposed to any chemicals from nearby tomato plants and weren't subsequently infested with whiteflies) produced about 140 nanograms of jasmonic acid per gram of dry weight and about 120 nanograms of salicylic acid per gram of dry weight, which indicates the baseline levels of these defensive chemicals in tomato plants. The graph also shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when subsequently infested by whiteflies, which is more than the amount produced by the control plants. This suggests that when infested with whiteflies, tomato plants that weren't previously exposed to air from nearby whitefly-infested plants begin producing higher levels of jasmonic acid to defend against the infestation. Finally, the graph also shows that plants exposed to air from whitefly-infested plants and then subsequently infested by whiteflies produced approximately 100 nanograms of jasmonic acid per gram of dry weight (which is less than control plants did) and approximately 175 nanograms of salicylic acid per gram of dry weight (which is more than control plants did). This suggests that whiteflies cause tomato plants to emit chemical signals that decrease nearby plants' defenses against whitefly herbivory relative to baseline levels and redirect their defenses toward microbial infection, thereby supporting the researchers' conclusion that whiteflies cause tomato plants to emit airborne chemical signals that induce nearby plants to preferentially defend against microbes even when experiencing herbivory from whiteflies.

Choice B is incorrect. Although the graph shows that tomato plants exposed to whitefly-infested plants produced less jasmonic acid than salicylic acid (about 100 nanograms of jasmonic acid per gram of dry weight vs. about 175 nanograms of salicylic acid per gram of dry weight), and although this might suggest that those plants are therefore better defended against microbial attack than against herbivory, this wouldn't be sufficient to conclude that this is the case:

without reference to the relative levels of defensive chemicals produced by the plants in the other experimental conditions, it would be impossible to conclude that exposure to air from whitefly-infested plants either induces or diminishes these plants' defenses to different kinds of threats. Furthermore, the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight and about 130 nanograms of salicylic acid per gram of dry weight, not that they produced the same amount of the two acids. Choice C is incorrect because it inaccurately describes the data in the graph. The graph shows that plants in all three conditions produced both jasmonic and salicylic acid, not that plants exposed to air from whitefly-infested plants produce no jasmonic acid at all. Choice D is incorrect. Although the graph shows that plants exposed to air from whitefly-infested plants produced about 100 nanograms of jasmonic acid per gram of dry weight when infested, which is less than the amount produced by control plants (about 140 nanograms per gram of dry weight), and although the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when infested (which is more than the amount produced by control plants) and about 130 nanograms of salicylic acid per gram of dry weight (which is more than the approximately 120 nanograms per gram of dry weight produced by control plants), this wouldn't support the researchers' conclusion that plants exposed to air from whitefly-infested plants preferentially defend against microbial infection—that is, that they increase their levels of salicylic acid and decrease their levels of jasmonic acid relative to control plants. Rather, the data would suggest that exposure to air from whitefly-infested plants reduces plants' defenses against herbivory, but these data wouldn't indicate that whiteflies induce plants to redirect their defenses to guard against microbial infection. Furthermore, the data concerning plants exposed to air from whitefly-free plants merely indicate that these plants increase their defenses against both herbivory and microbial infection when infested by whiteflies.

Q3

A

- A. photovoltaic cells with both silicon and perovskite are more efficient because they make use of more of the solar spectrum.
-

Choice A is the best answer because it presents the conclusion that most logically completes the text's discussion of the efficiency of various photovoltaic cells. The text indicates that Miranda-Pérez and her team found that adding a layer of perovskite to silicon-based cells increased the percentage of sunlight converted to usable electricity relative to the percentage converted to electricity by silicon-based cells. Using cells with only perovskite and no silicon, however, resulted in no improvement in efficiency over silicon-based cells. The text also states that perovskite captures light in the blue range of the solar spectrum and silicon captures light in the red range of the spectrum. If perovskite and silicon capture different parts of the solar spectrum and the combination of perovskite and silicon results in greater efficiency than the use of either perovskite or silicon alone does, then it is reasonable to infer that the improved efficiency of perovskite-silicon cells could be attributable to their making use of more of the solar spectrum than perovskite-based or silicon-based cells do.

Choice B is incorrect because it is directly contradicted by the text, which says that silicon-based cells convert 18%–22% of the sunlight that reaches them to electricity and that cells with only perovskite are no more efficient than silicon-based cells are. Perovskite-only cells thus must convert 22% or less—not more than 29.5%—of the sunlight that reaches them to electricity. Choice C is incorrect because the text focuses on a possible improvement to photovoltaic cells and makes no mention of any alternatives to those cells, so there is no reason to conclude that photovoltaic cells must be replaced with some other technology. Additionally, the text presents solar power as something that currently exists, so it wouldn't make sense to conclude that solar power will remain elusive. Choice D is incorrect because nothing in the text suggests that there is a need to evaluate the effectiveness of other minerals than perovskite. In fact, the text gives no indication that it is even possible to use other minerals than silicon and perovskite in photovoltaic cells. Instead, the text is focused on the effect of combining perovskite and silicon and the different parts of the solar spectrum that perovskite and silicon capture.

Q4

C

- C. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.
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Choice C is the best answer because it presents a finding that, if true, would most directly support the scholars' claim about the board game. The text explains that the remains of a 4,000-year-old board game were found in Oman. The text then states that scholars claim this board game was played mostly by traders. If the other known examples of this board game from the same time period were discovered along routes that seem to have been used primarily by traders, this finding would directly support the scholars' claim because it suggests that the game was largely played by traders who brought it with them for entertainment as they traveled.

Choice A is incorrect because this finding would suggest only that a single trader may have possessed examples of the board game, perhaps for the purpose of trading or selling the game to residents of Oman. For this reason, the finding wouldn't directly support the scholars' claim that the majority of the game's players were traders. Choice B is incorrect because this finding doesn't mention the board game at all, referring only to similar games found at other sites, and would therefore provide no direct support for the scholars' claim about the board game. Choice D is incorrect because this finding doesn't mention the board game at all, referring only to the remains of other goods found at the site in Oman, and would therefore provide no direct support for the scholars' claim about the board game.

Q5

B

- B. keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but decrease their weight.
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Choice B is the best answer because it most logically completes the text's discussion of Sidonian coins. As the text explains, researchers determined that Sidonian coins were made of silver and copper and that from 450 BCE to 367 BCE, the percentage of silver in each coin decreased from 98% to 74.2% while the percentage of copper increased from 1% to 24.7%. The text indicates that because the coins containing less than 80% silver weren't considered suitable for trade (suggesting that copper was less valuable than silver) and looked different from coins containing more silver, the researchers suspect there was a serious loss in confidence in the currency in Sidon in 367 BCE when the copper content was high. It's reasonable to assume that it wasn't possible to boost confidence simply by devoting a greater amount of valuable silver to the currency, since Sidon was under significant and ongoing financial pressure; however, keeping the total amount of silver the same and reducing the amount of copper in the coins would have resulted in smaller coins with a higher percentage of silver. Therefore, it makes sense to suggest that Abd'astart I (the ruler after 367 BCE) likely restored confidence in the currency by deciding to keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but to decrease the coins' weight.

Choice A is incorrect because the text conveys that a crisis in confidence in the currency of Sidon likely occurred around 367 BCE because the percentage of silver in coins had fallen below 80% (presumably because Sidon's financial pressures meant that less silver was available for currency), making the coins unsuitable for trade. Thus, announcing that the threshold for the percentage of silver in coins would be raised—that is, that coins would need to contain even more than 80% silver to be suitable for trade—likely would have worsened the crisis rather than relieved it. Choice C is incorrect because the text strongly suggests that a crisis in confidence in the currency of Sidon was caused by the proportion of silver to copper in the coins in 367 BCE, with 74.2% being too little silver for the coins to be considered suitable for trade; therefore, it's unlikely that minting coins with a similar proportion of silver to copper (that is, still around 74.2% silver) would have restored confidence, even if the coins were heavier. Choice D is incorrect because the text gives no indication that funding the mining of more copper would have relieved a crisis in confidence in the currency of Sidon. The text establishes that Sidonian coins that visibly contained copper weren't considered suitable for trade, so Abd'astart I wouldn't have wanted to add even more copper to them, and it's unclear how else copper mining would affect views of the currency.

Q6

A

- A. Analysis of the animal bones showed that the cattle's diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.
-

Choice A is the best answer because it presents a finding that, if true, would most strongly support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals' domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team's analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle's diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team's conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep's and goats' diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team's conclusion that cattle were likely raised closer to those settlements than sheep and goats were. Choice C is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn't support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without access to human-grown crops. Hence, even if cattle's diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. Choice D is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements' crops.

Q7

A

- A. across cultural contexts, people who share similar profiles of personality traits tend to prefer listening to similar types of Western music.
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Choice A is the best answer because it most logically completes the text's discussion of the cross-cultural study on personality traits and music preferences. The text first explains that previous studies in individual Western societies had found links between personality traits and five musical factors that shape individual preferences for Western music styles. This finding suggests that people in those societies who have similar personality traits likely prefer listening to similar types of Western music. The text goes on to state that Greenberg et al. conducted a study that expanded this research to a broader, cross-cultural context (fifty-three countries across six continents). This study validated that the five-factor framework accurately represented participants' preferences for Western music and that the study found patterns linking those preferences to personality traits. It follows, then, that people with similar profiles of personality traits tend to prefer similar types of Western music across cultural contexts.

Choice B is incorrect because Greenberg et al.'s study examined only preferences for Western music styles, so the text provides no basis for inferring that the five-factor framework could predict preferences for non-Western music styles. The study was specifically limited to Western musical styles, and extending the study's findings to non-Western music would go beyond what the study investigated or supports. Choice C is incorrect because the text doesn't mention what researchers had previously assumed about cultural variation in the strength of the relationship between personality traits and musical preferences. The text presents the cross-cultural consistency of the association between musical preferences and personality traits as a new finding but doesn't mention previous expectations or assumptions about this association. Choice D is incorrect because the text doesn't address familiarity with Western music styles or suggest that familiarity influences preference strength for those styles. Greenberg et al.'s study focused on the relationship between personality traits and musical preferences across cultural contexts, not on how exposure or familiarity might affect those preferences.

Q8

C

- C. Evidence emerges that the amount of iron in some rocky planets is considerably higher than the amount in their host star.
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Choice C is the best answer because it presents a finding that, if true, would weaken the astronomers' claim about the makeup of host stars and their planets. The text explains that because stars and planets begin forming from the same gas and dust, astronomers believe planets should be composed of the same materials as their host stars, but in equal or smaller quantities. The finding that the amount of iron in some rocky planets is much higher than the amount in their host star would weaken the astronomers' claim because it would show that some planets contain the same material as their host star, but in higher quantities.

Choice A is incorrect because a finding only about the makeup of stars, whether they've cooled or not, would provide no information about the makeup of planets. Thus, it wouldn't have any bearing on the claim that planets and their host stars are composed of the same materials in differing quantities. Choice B is incorrect because a finding about two host stars having similar proportions of certain materials wouldn't provide any information about the makeup of planets. Thus, it wouldn't be relevant to the claim that planets and their host stars are composed of the same materials in differing quantities. Choice D is incorrect because the text indicates that the astronomers' claim is based on a fact—that stars and planets begin forming from the same gas and dust in space—which would remain true regardless of the effectiveness of a method for analysis of compositions. The text does cite analysis of rocky planets in our solar system and the Sun, but only as a single piece of evidence that is consistent with the claim and not as the source of the claim; the finding that the method used for that analysis is less effective in other scenarios wouldn't weaken a claim that's based on knowledge of how stars and planets initially form.

Q9

A

- A. the Aztecs would have disputed the idea that the morality of an individual's actions can be assessed by appealing to standards of behavior that are independent of the individual's social circumstances.
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Choice A is the best answer because it most logically completes the text's discussion about Aztec (Nahua) ethics. The text indicates that, according to Purcell's interpretation of available Aztec philosophical works, the Aztec ethical system views an individual's actions in relation to that individual's societal role and how the actions affect the community. The text contrasts this view with another held by some ethicists, namely that actions are morally good or bad regardless of the context in which they occur. Thus, Purcell's analysis suggests that the Aztecs would have asserted that the morality of an individual's actions are rooted in that person's position in the community and the actions' effects and therefore cannot be determined in the absence of that context.

Choice B is incorrect. Although the text indicates that morally judging an action according to Aztec ethics requires an understanding of the action's effects and the individual's social circumstances, it does not specify that only members of that society can acquire this information. Choice C is incorrect because it implies that the Aztecs considered some actions good or bad regardless of the surrounding context, which contradicts the text's claim that the Aztecs believed that the morality of an individual's action is dependent on the action's effects on the community and the person's specific circumstances. Choice D is incorrect. Although the text indicates that in Aztec ethics the morality of an action depends in part on how it affects the community, this is only one of the two factors—the other being the person's societal role—that need to be considered. Therefore, it is possible that two actions with the same effect on the community could be considered morally distinct if they are performed by individuals in different social roles.

Q10

C

- C. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (*Chaetostoma yurubiense*) bumped into obstructions more often than the narrower-headed dusky smooth-hound (*Mustelus canis*) did.

Choice C is the best answer because it presents a finding that, if true, would most directly support the second research team's hypothesis about how a fish's head shape affects the fish's ability to detect obstructions. The text explains that fish detect obstructions using water vortices and that testing by Yanagisuru, Akanyeti, and Liao showed that fish with wide heads struggle to distinguish medium-sized vortices from general turbulence in the water. Based on this finding, a second research team hypothesized that in low-visibility conditions, wider-headed fish would be less effective at detecting obstructions than narrower-headed fish. The finding that wider-headed bristlemouth fish bumped into obstructions more often than narrower-headed dusky smooth-hound fish did under the specified conditions (low visibility with medium-sized vortices) would directly support the second team's hypothesis.

Choice A is incorrect because it would provide only partial support for the second research team's hypothesis. While this finding shows that wide-headed bristlemouth fish bumped into many obstructions, it provides no comparison with narrow-headed fish. The hypothesis specifically predicts a difference in detection ability between fish with wider heads and those with narrower heads, so supporting evidence would need to include data from both head types to allow for this comparison. Choice B is incorrect because it would not support the second research team's hypothesis comparing obstruction detection between fish with wide heads and those with narrow heads. Instead, it examines variation within a single narrow-headed species, which would not address the hypothesis about differences between fish with wider and narrower head shapes. Choice D is incorrect because this finding would contradict rather than support the second research team's hypothesis. The hypothesis predicts that wider-headed fish will be less likely to detect obstructions than narrower-headed fish in low-visibility conditions, which would result in wider-headed fish bumping into obstructions more frequently. A finding showing equal bumping rates between fish with the two head shapes would indicate no relationship between head width and detection ability, directly contradicting the hypothesis.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

